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Documentation of Course Work
on SQL & Data Processing

Vilnius, 2026

Topic: Online Book Store

A database for an online book store has been developed. The project includes OLTP and OLAP databases, an ETL process, analytical queries, and a Power BI dashboard.

1. OLTP Database

Purpose: Storing operational information about books, authors, customers, orders, and payments.

What has been done:

- 10 tables in 3NF
- Primary and foreign keys, constraints, indexes
- Schema in `schema.dbml`

Tables: Customers, Authors, Publishers, Categories, Books, BookAuthors, Orders, OrderItems, Payments, Reviews.

Data:

- 5 CSV files without surrogate keys
- `load_data.sql` script loads data with duplicate checking (can be run multiple times)

2. OLAP Data Warehouse

Purpose: Analytics and reporting.

What has been done:

- Snowflake schema
- 2 fact tables: FactSales, FactPayments
- SCD Type 2 in DimCustomer (stores change history)
- BridgeBookAuthor for linking books and authors
- OLAP is separated from OLTP

Dimensions: DimCustomer, DimBook, DimDate

3. ETL процесс

The `etl_to_dwh.sql` script transfers data from OLTP to OLAP:

- Populates DimDate from order dates
- Updates DimCustomer with history (SCD Type 2)
- Loads DimBook with denormalized categories
- Populates fact tables and bridge table
- All inserts use existence checks – no duplicates

4. Запросы

OLTP (3 queries):

1. Top selling books by year
2. Revenue by category
3. Top spending customers

OLAP (3 queries):

1. Top selling books by year
2. Revenue by category
3. Top spending customers (current only)

5. Power BI

The dashboard connects to OLAP and contains:

- Title: "Online Book Store Analytics"
- 2 slicers: Year, Category
- Created 3 visualizations to display information.

6. Conclusion

During this course project, a complete data storage and analytical system for an online book store was designed and implemented.

Key results:

- The OLTP database enables efficient operational record-keeping: storing information about books, authors, customers, orders, and payments. Normalization and indexes ensure data integrity and transaction speed.
- The OLAP data warehouse provides the ability to analyze business metrics: identify best-selling books, track revenue by category, and identify top customers. SCD Type 2 allows viewing the history of customer data changes.
- The ETL process automatically transfers data between systems without duplication, ensuring up-to-date analytics.
- The Power BI dashboard provides a visual representation of key metrics, enabling data-driven management decisions.